

ДЗ 7

Вар 21

№	A	B
21.	6,5	0,04

Задание 1

Формат Ф1

$$A = (6.5)_{10} = (6.8)_{16} = (0.68)_{16} * 16^1$$

0	1	0	0	0	0	0	1	0	1	1	0	1	0	0	0	0	0	0	0
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$$B = (0.04)_{10} = (0.0(A3D70))_{16} = (0,(A3D70))_{16} * 16^{-1}$$

MB

0	0	1	1	1	1	1	1	1	0	1	0	0	0	1	1	1	1	1	0
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1 округление
к ближайшему

$$\text{Sign}C = \text{Sign}A \oplus \text{Sign}B = 0 \oplus 0 = 0.$$

$$X_A = P_A + d; \quad X_B = P_B + d;$$

$$X_C = X_A + X_B - d;$$

$$\begin{array}{r}
 X_A = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \\
 + \\
 X_B = 0 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \\
 \hline
 X_A + X_B = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\
 - \\
 d = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\
 \hline
 X_C = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0
 \end{array}$$

$$P_C = 0.$$

№	Операнды	СЧП (старшие разряды)	В/СЧП (младшие разряды)	Признак коррекции
0	СЧП	0000000000000000	1010001111100	0
1	[2M _A] _{пр}	00011010000000	2M _A	0
	СЧП	00011010000000	1010001111100	
	СЧП->2	00000110100000	001010001111	
2	[M _A] _{доп}	11110011000000	-M _A	1
	СЧП	11110011000000	001010001111	
	СЧП->2	11111100110000	000010100011	
3	0		-	1
	СЧП	11111100110000	000010100011	
	СЧП->2	11111111001100	000000101000	
4	[M _A] _{пр}	00001101000000	M _A	0
	СЧП	00001100100110	000000101000	
	СЧП->2	00000011001001	100000000101	
5	[2M _A] _{пр}	00011010000000	2M _A	0
	СЧП	00011101001001	100000000101	
	СЧП->2	00000111010101	110000000010	
6	[2M _A] _{пр}	00011010000000	2M _A	0
	СЧП	00100001010101	110000000010	
	СЧП->2	00001000010100	100110000000	
7	СЧП	00001000010100	100110000000	0
	M _C	00001000010100	100110000000	

$$C^* = (0.429)_{16} \cdot 16^0 = 0.2600098$$

$$\delta C = \left| \frac{\Delta C}{C_T} \right| \cdot 100\% = \left| \frac{-0.0000098}{0,26} \right| \cdot 100\% = 0,004\%$$

- неточным представлением операндов;
- потерей младших бит СЧП.

$$A = (6.5)_{10} = (6.8)_{16} = (0.1101000)_2 * 2^3$$

0	1	0	0	0	0	0	1	1	1	0	1	0	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

0	0	1	1	1	1	1	0	0	0	1	0	0	0	1	1	1	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$X_C = X_A + X_B - d;$$

$$\begin{array}{r} X_A = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \\ + \\ \underline{X_B = 0 \ 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0} \end{array}$$

$$X_A + X_B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{d} = \begin{array}{cccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$$X_C = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$P_C = -1.$$

№ ы	Операнд	СЧП (старшие разряды)	В/СЧП (младшие разряды)	Признак коррекции
0	СЧП	00000000000000000000	101000111101	0
	$[4M_A]_{\text{доп}}$	11100110000000000000	-4M _A M _A	
	$[M_A]_{\text{пр}}$	00000110100000000000		
1	СЧП	11101100100000000000	101000111101	1
	СЧП->4	11111110110010000000	0000101000011	
	$[4M_A]_{\text{пр}}$	00011010000000000000	4M _A -M _A	
	$[M_A]_{\text{доп}}$	11111001100000000000		
2	СЧП	00010010010010000000	0000101000011	0
	СЧП->4	00000001001001001001	0000000001010	
	$[8M_A]_{\text{пр}}$	00110100000000000000	8M _A 2M _A	
	$[2M_A]_{\text{пр}}$	00001101000000000000		
3	СЧП	01000010001001001001	0000000001010	0
	СЧП->4	000001000010001001	1001000000000	
	СЧП	000001000010001001	1001000000000	

$$C^* = (0.1000010001)_2 \cdot 2^{-1} = (0.0100010001)_2 = 0.2583.$$

$$\Delta C = C_T - C^* = 0.26 - 0.2583 = 0,0017,$$

$$\delta C = \left| \frac{\Delta C}{C_T} \right| \cdot 100\% = \left| \frac{0,0017}{0.26} \right| \cdot 100\% = 0.654\%.$$

Погрешность результата вызвана неточным представлением операндов и потерей младших бит СЧП.

В формате Ф2 погрешность больше из-за более крупного шага умножения.